Yahya's Cyclic Prime Residue Framework

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Abstract

This paper presents an observation regarding the distribution of prime numbers when mapped onto a repeating seven-letter cyclic labeling system [cite: 1]. By labeling integers with a periodic sequence of seven symbols (A–G), where each symbol corresponds to a residue class modulo 7, a notable variance in the distribution of prime numbers arises across successive seven-integer blocks [cite: 1]. The pattern suggests that primes exhibit non-uniform clustering across residue classes in locally bounded ranges, while maintaining long-term equidistribution statistically [cite: 1]. This paper outlines the observed pattern, formalizes the structure behind it, and provides theoretical context based on modular arithmetic [cite: 1].

1 Introduction

Prime numbers play a central role in number theory, exhibiting both deterministic structure and apparent randomness [cite: 1]. The classical approach to understanding primes frequently involves modular arithmetic, in which integers are classified according to their residue modulo a fixed integer [cite: 1]. This paper examines the distribution of prime numbers under modulo 7, using a cyclic labeling system (A–G) that repeats every seven integers [cite: 1].

For the purposes of early pattern recognition, the integer 1 was initially treated as prime [cite: 1]. However, this paper proceeds with the standard mathematical convention that 1 is not a prime number [cite: 1].

2 Definition of the Labeling System

Let the letters A, B, C, D, E, F, G correspond to the integers 1 through 7 respectively [cite: 1]. For any positive integer n, we define its label by [cite: 1]:

$$L(n) = \text{the letter corresponding to } ((n-1) \pmod 7) + 1.$$

Thus, the sequence progresses [cite: 1]:

$$\cdot 1 \to A, \quad 2 \to B, \quad 3 \to C, \quad 4 \to D, \quad 5 \to E, \quad 6 \to F, \quad 7 \to G, \quad 8 \to A, \quad 9 \to B, \quad \ldots$$

and so forth [cite: 1]. This creates a periodic mapping of period 7 [cite: 1].

3 Observed Pattern in Local Blocks

When examining integers in consecutive blocks of seven, the distribution of primes per block shows noticeable variation [cite: 1]. For example [cite: 1]:

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- In some blocks, primes may cluster at labels A, B, or C [cite: 1].
- In subsequent blocks, primes may instead cluster more strongly at labels D or F [cite: 1].

This variation suggests that local distributions differ significantly across blocks, even though the labeling system itself is fixed [cite: 1].

4 Theoretical Explanation

All primes greater than 7 must be congruent to one ofneat the residues [cite: 1]:

$$1, 2, 3, 4, 5, 6 \pmod 7,$$

since any integer congruent to 0 (mod 7) would be divisible by 7 [cite: 1]. Thus, no residue class is theoretically forbidden except 0 (mod 7) [cite: 1]. Long-term analytic number theory (e.g., Dirichlet's theorem on arithmetic progressions) states that primes tend toward statistical equidistribution among the allowed residue classes [cite: 1]. However, in locally bounded regions (such as a single block of seven integers), the distribution is influenced by small-scale irregularities in prime occurrence [cite: 1]. These local fluctuations produce the appearance of shifting patterns between blocks [cite: 1].

5 Conclusion

The observed prime-label distribution variation is a manifestation of short-range irregularities in prime placement interacting with a fixed-period labeling system [cite: 1]. While no fixed deterministic pattern repeats indefinitely, the observation highlights a meaningful way to visualize prime distribution behavior on modular cycles [cite: 1].

Future work may include [cite: 1]:

- Extending computational verification to larger ranges [cite: 1].
- Graphing prime density per residue class across thousands of blocks [cite: 1].
- Investigating whether similar patterns emerge under moduli other than 7 [cite: 1].

Acknowledgment

This framework originated from independent analytical exploration by the author at age 14 [cite: 1].